

Assignment 8

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2025-11-30

```
library(readr)
library(dplyr)

Indicators_of_Anxiety_or_Depression_Based_on_Reported_Frequency_of_Symptoms_During_Last_7_Days <-
  read_csv("Indicators_of_Anxiety_or_Depression_Based_on_Reported_Frequency_of_Symptoms_During_Last_7_Days.csv")

colSums(is.na(Indicators_of_Anxiety_or_Depression_Based_on_Reported_Frequency_of_Symptoms_During_Last_7_Days))

##           Indicator           Group           State
##           0           0           0
##      Subgroup           Phase           Time Period
##           0           0           0
## Time Period Label Time Period Start Date   Time Period End Date
##           0           0           0
##           Value           Low CI           High CI
##           540           540           540
## Confidence Interval           Quartile Range
##           540           3834

indicators <- Indicators_of_Anxiety_or_Depression_Based_on_Reported_Frequency_of_Symptoms_During_Last_7_Days
filter(!is.na(Value) & !is.na(`Low CI`) & !is.na(`High CI`))

library(dplyr)
library(ggstatsplot)

## You can cite this package as:
## Patil, I. (2021). Visualizations with statistical details: The 'ggstatsplot' approach.
## Journal of Open Source Software, 6(61), 3167, doi:10.21105/joss.03167

library(ggplot2)

sex_data <- indicators %>%
  filter(
    Group == "By Sex",
    Subgroup %in% c("Male", "Female"),
    Indicator %in% c(
      "Symptoms of Anxiety Disorder",
      "Symptoms of Depressive Disorder"
    )
  ) %>%
  mutate(
    Subgroup = factor(Subgroup, levels = c("Male", "Female"))
  )
```

```

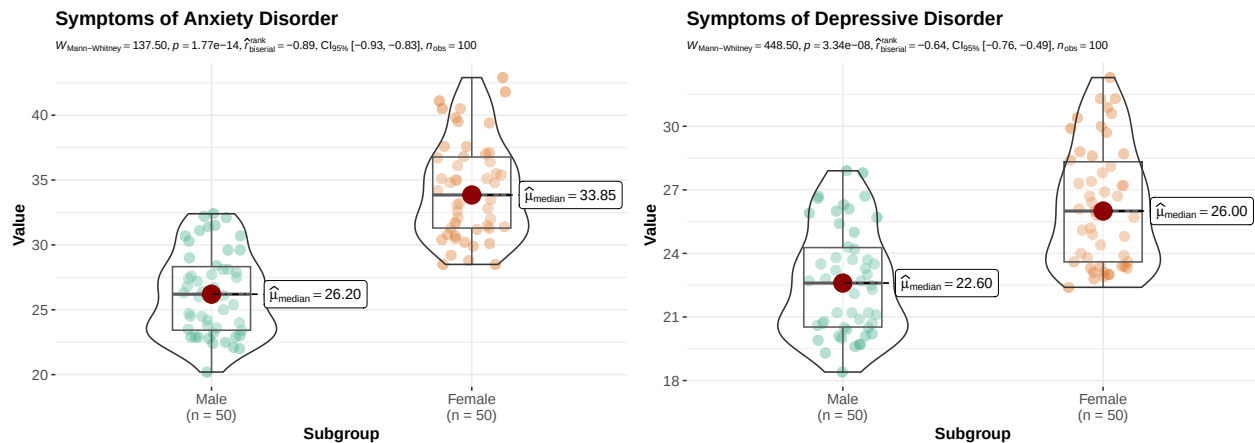
ggstatsplot::grouped_ggbetweenstats(
  data      = sex_data,
  x         = Subgroup,
  y         = Value,
  grouping.var = Indicator,

  type      = "nonparametric",

  centrality.type = "nonparametric",

  digits      = 2,
  bf.message  = FALSE,
  plotgrid.args = list(nrow = 1),
  ggplot.component = list(
    theme(
      plot.subtitle = element_text(size = 7),
      strip.text    = element_text(size = 10),
      axis.text.x   = element_text(size = 9),
      axis.text.y   = element_text(size = 9),
      axis.title    = element_text(size = 10)
    )
  )
)

```



CONCLUSION: When the impact of gender on anxiety and depression symptom levels is examined, it is observed that females report higher symptom values than males for both disorders. In the context of anxiety, the median value for females is approximately 33.85, whereas for males, it stands at around 26.20. Violin plots indicate that the symptom distribution among female participants is both wider and extends toward higher values, revealing greater variance and symptom intensity in women. Mann–Whitney test results strongly corroborate this observation: a p -value of $1.77\text{e-}14$ demonstrates that the difference is statistically highly significant, while a rank-biserial correlation of -0.89 indicates a very large effect size. Thus, gender creates a distinct and robust separation in anxiety symptoms. A similar pattern is observed for depression symptoms, although the effect size is relatively smaller. The median for females is 26.00, compared to approximately 22.60 for males. The symptom distribution for females again spans a wider range and leans toward higher values. While the Mann–Whitney test p -value of $3.34\text{e-}08$ confirms the statistical significance of this difference, the rank-biserial effect of -0.64 suggests a medium-to-high level difference, albeit not as strong as in anxiety. In conclusion, while females report higher symptom levels in both disorders, the gender disparity is notably more pronounced in anxiety.

```

library(ggplot2)
library(ggribes)
library(viridis)
library(dplyr)
library(readr)
library(forcats)
library(tidyr)

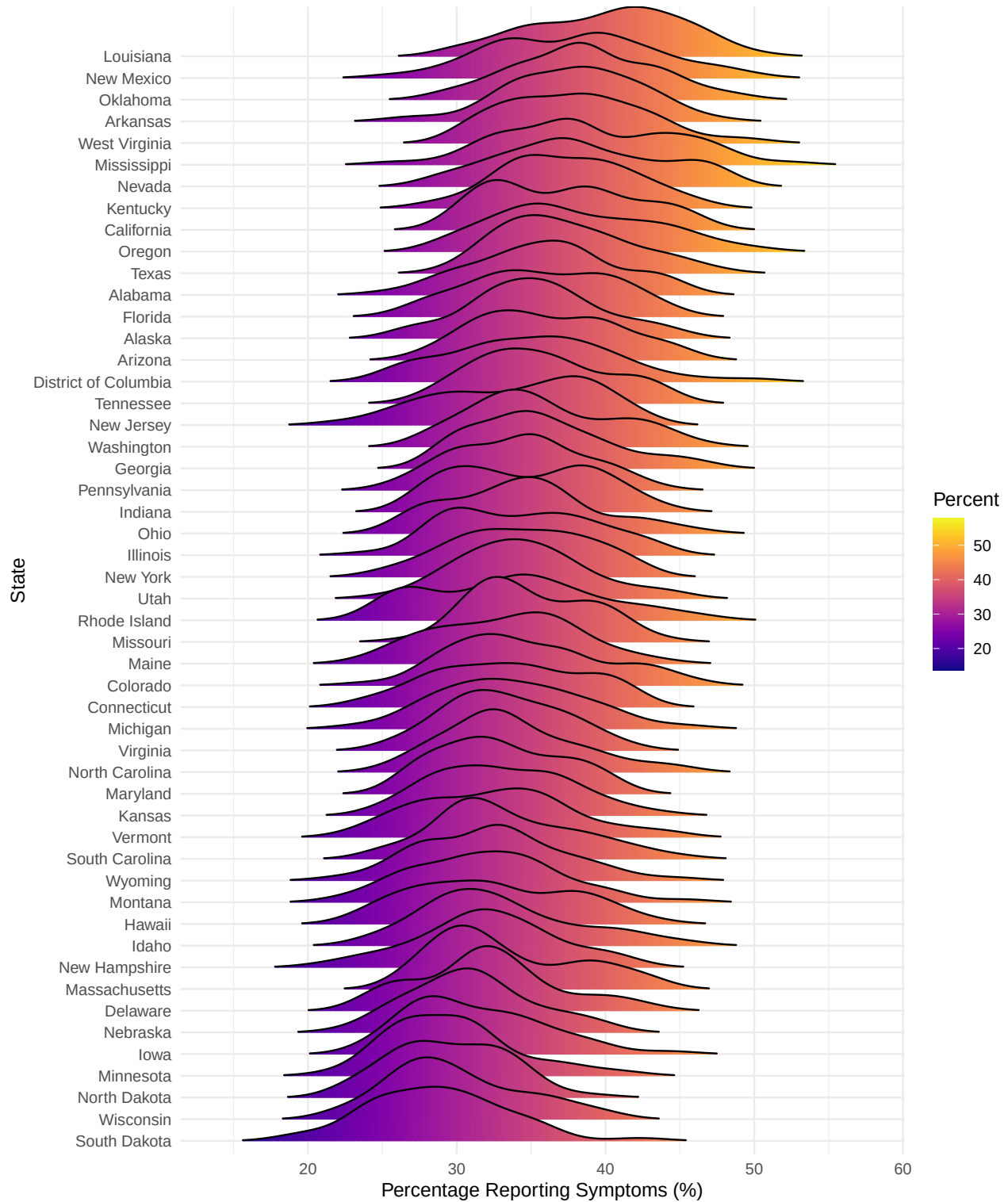
state_data <- indicators %>%
  filter(
    Group == "By State",
    Indicator == "Symptoms of Anxiety Disorder or Depressive Disorder"
  ) %>%
  mutate(State = Subgroup) %>%
  drop_na(Value)

state_all_data <- state_data %>%
  mutate(State = fct_reorder(State, Value, .fun = median))

ggplot(
  state_all_data,
  aes(
    x = Value,
    y = State,
    fill = after_stat(x)
  )
) +
  geom_density_ridges_gradient(
    scale = 3,
    rel_min_height = 0.01
  ) +
  scale_fill_viridis(
    name = "Percent",
    option = "C"
  ) +
  labs(
    title = "Distribution of Anxiety/Depression Symptom Rates Across States",
    subtitle = "All states (By State, Symptoms of Anxiety or Depressive Disorder)",
    x = "Percentage Reporting Symptoms (%)",
    y = "State"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    legend.position = "right",
    panel.spacing = unit(0.15, "lines")
  )

```

Distribution of Anxiety/Depression Symptom Rates Across States
All states (By State, Symptoms of Anxiety or Depressive Disorder)



CONCLUSION: This ridgeline graph shows the trajectory of anxiety or current symptoms across all states in the US. The graph represents a curve, representing the occurrence of symptoms observed in a given state over time. The horizontal axis shows the percentage of people reporting symptoms, and the vertical axis shows the states. The state names are not randomly listed, but rather are ordered according to the median

symptoms calculated for the state; states with lower medians appear lower on the list, and states with higher medians appear higher. This listing makes the comparison between states more visually understandable. The color scale also aids in interpreting the graph: purple tones represent lower percentages, while shades toward yellow and orange represent higher percentages. In some states (Louisiana, New Mexico, and Arizona), the curves extend further to the right, with warmer colors (yellow and orange) covering a larger area, suggesting that symptom rates are both higher and more frequent in these states. Conversely, in some states, such as Minnesota, Wisconsin, and North Dakota, the distributions are narrower and centered around lower values, suggesting that symptom rates are relatively lower and more stable in these states. Overall, the graph reveals significant differences in both symptom levels and the breadth of the distribution across states, and that mental health indicators have been quite geographically heterogeneous during the pandemic.

NOTE: Actually, a map-style visualization could have been made here instead of a ridgeline chart. I preferred the ridgeline chart because I wanted to be more comparison and distribution-oriented.

```
library(forcats)
library(ggplot2)
library(ggthemes)
age_data <- indicators %>%
  filter(
    Group == "By Age",
    Indicator %in% c(
      "Symptoms of Anxiety Disorder",
      "Symptoms of Depressive Disorder",
      "Symptoms of Anxiety Disorder or Depressive Disorder"
    )
  ) %>%
  drop_na(Value)

age_summary <- age_data %>%
  group_by(Age = Subgroup, Indicator) %>%
  summarise(mean_value = mean(Value, na.rm = TRUE), .groups = "drop") %>%
  mutate(
    Age = fct_reorder(Age, mean_value)
  )

ggplot(age_summary,
  aes(x = Age, y = mean_value, fill = Indicator)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    x = "",
    y = "Average percentage (%)",
    fill = "Indicator",
    title = "Comparison of Anxiety and Depression Symptom Prevalence Across Age Groups",
    subtitle = "Average Self-Reported Symptom Prevalence Over All Survey Waves"
  ) +
  scale_y_continuous(limits = c(0, 60)) +
  scale_fill_manual(values = c(
    "Symptoms of Anxiety Disorder" = "grey70",
    "Symptoms of Depressive Disorder" = "grey40",
    "Symptoms of Anxiety Disorder or Depressive Disorder" = "lightblue"
  )) +
  theme_hc() +
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5),
```

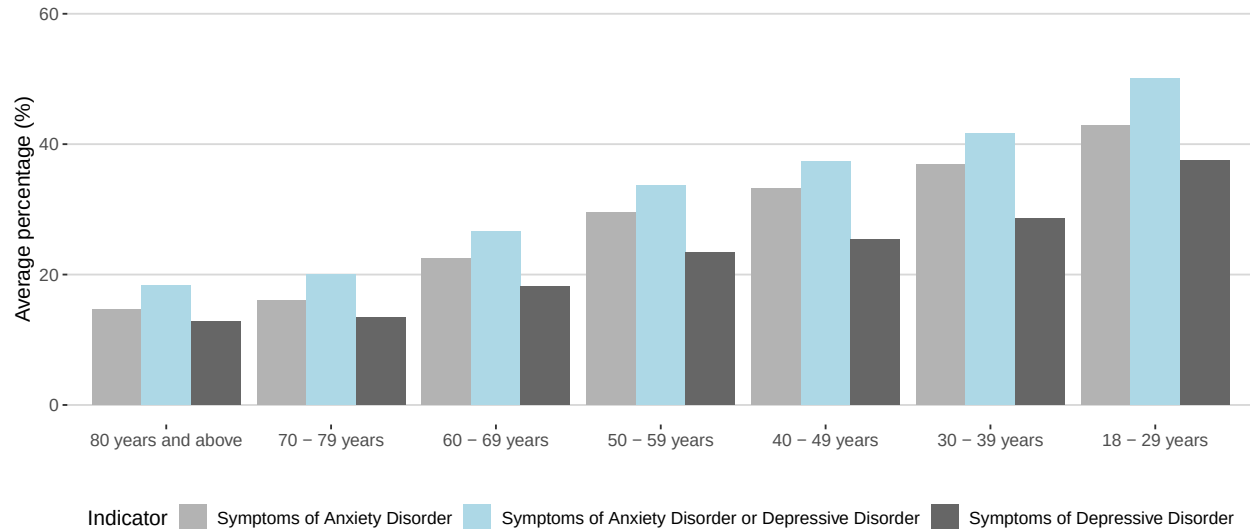
```

plot.title = element_text(face = "bold", size = 14),
legend.position = "bottom"
)

```

Comparison of Anxiety and Depression Symptom Prevalence Across Age Groups

Average Self-Reported Symptom Prevalence Over All Survey Waves



CONCLUSION: The graph compares the average prevalence rates of anxiety and depression across different age groups. Generally, the highest symptom levels are observed between ages 18 and 29. This cohort exhibits a higher prevalence of symptoms than other age groups, and the composite index (“anxiety or depression”) reaches its peak within this demographic. This indicates that young adults report mental health issues more frequently. There is a steady decline in symptom rates as age increases. For instance, while rates remain elevated in the 30–39 and 40–49 age groups, they are lower than in younger adults. In the 60-and-older group, rates of both anxiety and depression decrease significantly, with the lowest rates seen in individuals aged 80 and over. Furthermore, the combined rate of “anxiety or depression” is consistently higher than the individual rates, suggesting that some individuals experience both symptoms simultaneously. Overall, the data demonstrates that mental health symptoms are particularly prevalent in younger age groups and diminish with age.